	Target engagement	Dose	Efficacy	Safety
#6	**Alzheimer disease** HRH3 antagonists (3)	**2014** Pig PET and in-vitro H3 affinity of GSK239512 (vs human [11C]GSK189254 PET)	**2014** Rat pup inhibitory avoidance Target exposure reached; ADAS-Cog failed	**2016** Rodent procognitive battery Three rodent successes, three AD failures	**2016** Rat safety, EEG; monkey tox Sponsor confirms dose-dependent sleep disturbance
	Alzheimer disease HTR6 antagonists (3)	**2023** Non-human primate PET 5-HT6 receptor occupancy △ Rat under-called brain penetration	**2022** Rat cognition with donepezil Rat efficacious exposure reached; trial failed	**2022** Rat cognition with donepezil Rat efficacy did not translate; Phase 2 null	**2014** IND-enabling toxicology (species and findings unpublished; no NDA, so no FDA review)
	Alzheimer disease anti-APP antibodies (3)	**2021** tg-ArcSwe mice, mAb158 Mice: no plaque fall; patients: PET fell	**2014** hAPP/PS1 mice, crenezumab 15 mg/kg failed; higher doses failed too	**2014** hAPP/PS1 mice, crenezumab Crenezumab missed both primaries	**2017** Aged APP mice given 2H6 (ponezumab parent) △ Excess microbleeds missed
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2019** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2013** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
	asthma ADRB2 agonists (4)	**2017** Carmoterol binding kinetics (guinea-pig trachea, bovine beta2, GppNHp shift), basis of long action	**2014** Guinea-pig/human bronchial onset and duration (AZD3199) △ 2 ug daily missed 24-h trough	**2014** Guinea-pig/human bronchial onset and duration (AZD3199) △ 24-h trough FEV1 primary missed	**2017** Guinea-pig atria and papillary muscle chronotropy/inotropy, cardiac side-effect model (carmoterol)
	asthma NR3C1 agonists (4)	**2017** Rat allergic airway model: lung oedema, thymic involution	**2020** Rat allergic airway model: lung oedema, thymic involution Top dose missed week-12 FEV1	**2014** Late asthmatic response to allergen in mild atopic asthmatics Half-ICS LAR effect; 1 mg trial failed	**2020** Rat allergic airway model: lung oedema, thymic involution
	asthma anti-IL13 antibodies (4)	**2013** IL-13-stimulated bronchial epithelial cells and patient brushings △ Airway eosinophils did not fall	**2018** Tralokinumab population PK/PD dose-exposure-FEV1 model Model dose reached; phase 3 failed	**2018** hIL-13 mice, tralokinumab Phase 3 exacerbation and OCS primaries missed	**2022** Cynomolgus repeat-dose toxicology (tralokinumab, lebrikizumab)
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) △ Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) △ Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays △ Missed liver injury and clots
	breast cancer ESR1 degrader (5)	**2026** ESR1-mutant MCF7-Y537S cells, Y537S and mutant-ER PDX	**2026** ESR1-mutant MCF7-Y537S cells, Y537S and mutant-ER PDX △ Target hit; amcenestrant failed	**2025** ESR1-mutant MCF7-Y537S cells, Y537S and mutant-ER PDX △ Amcenestrant fails PFS vs ET	**2026** Rat and dog repeat-dose toxicology (oral ER degraders) △ hERG, dog clean; patient QTc up
	breast cancer HSP90AA1/HSP90AB1 inhibitors (5)	**2014** HER2+ SKOV-3 xenograft, 89Zr-trastuzumab PET of luminespib HER2 downregulation	**2013** HER2+ breast xenografts	**2013** HER2+ breast xenografts HER2+ trial: 0 of 26 responses, stopped	**2013** Mouse xenografts, cardiac and liver readouts (ganetespib) △ Eye/heart DLTs missed; no renal
	breast cancer PI3K/MTOR inhibitors (5)	**2018** Novartis BEZ235 xenografts with ex-vivo tumour PI3K/Akt pathway readout	**2015** XL765 xenografts, pAKT PD MTD reached, phase II failed, abandoned	**2015** XL765 xenografts, pAKT PD No responses, PFS6 8% in phase II	**2016** GDC-0980 PI3K-activated xenografts, exposure-linked PD Hyperglycaemia DLT; model silent
	breast cancer anti-IGF1R antibodies (4)	**2011** MCF-7 mice, IMC-A12 △ GFS fell in under 40% predicted	**2015** MCF-7 mice, IMC-A12 Target trough reached; drug failed; abandoned	**2017** Tamoxifen-resistant MCF-7L cells and xenografts (dalotuzumab) △ Dalotuzumab + mTOR block: none	**2020** Monkey tox (teprotumumab) Monkeys missed hyperglycaemia (10% in TED)
	breast cancer PARP1/2 inhibitors (3)	**2021** Pamiparib BRCA1-mutant xenograft, PAR PD, mouse PK/PD	**2021** Pamiparib BRCA1-mutant xenograft, PAR PD, mouse PK/PD △ Target reached, potentiation failed	**2022** Pamiparib BRCA1-mutant xenograft, PAR PD, mouse PK/PD △ Veliparib + carboplatin: nothing	**2020** 2X-121 BRCA-deficient mice, PAR PD, chemo potentiation △ GEMM missed blood toxicity
	breast cancer PI3K (pan) inhibitors (3)	**2014** PI3K-activated lines and mice	**2016** PI3K-activated lines and mice Target exposure reached, pictilisib failed	**2016** Breast PDX, ex vivo screen PIK3CA-mutant enrichment not seen in FERGI	**2017** PI3K-activated lines and mice Liver toxicity unflagged by preclinical package
	breast cancer PIK3CA inhibitors (3)	**2017** PIK3CA-mutant cell-line panel (CCLE) and PIK3CA-mutant breast cancer xenografts	**2019** PIK3CA-mutant cell-line panel (CCLE) and PIK3CA-mutant breast cancer xenografts	**2020** Index patient's PTEN-null, BYL719-refractory metastasis PDX; PTEN-knockdown PIK3CA mutants	**2024** Secondary pharmacology and tox: mouse insulin/glucose tolerance, rat and dog repeat-dose studies
	breast cancer pan-HER inhibitors (3)	**2014** Tamoxifen-resistant MCF7 and T47D lines and xenografts under tamoxifen or fulvestrant	**2017** HER2+ breast lines and mice	**2018** HER2+ breast lines and mice	**2017** HER2+ breast lines and mice Diarrhoea DLT despite stated animal window
#9	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cynomolgus endotoxin-induced TNF-alpha PD model (PH-797804)	**2012** Rodent rheumatoid-arthritis models, ex-vivo LPS whole-blood cytokine PD (ARRY-797)	**2014** Rat repeat-dose intra-articular toxicology (FX005)
	osteoarthritis NTRK1 inhibitors (3)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat PGPS knee arthritis flare (GZ389988, intra-articular) △ KO: TrkA block enough; it wasn't	**2021** Rat 4- and 13-week oral neurotoxicity (ASP7962) △ Rat silent; joint inflammation in 2/3
#10	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2017** Human colorectal patient-derived xenografts (PDX) in mice	**2023** Dog CV safety, rat/dog tox Dogs silent; 14% grade 3-4 hypertension
	colorectal cancer VEGFR multikinase inhibitors (3)	**2009** VEGF-induced uterine edema in mice (vascular-permeability PD model, ABT-869/linifanib)	**2014** GastroPlus PBPK of famitinib, rat and monkey	**2017** Human tumour cell-line xenografts in mice incl. colon carcinoma (ABT-869/linifanib)	**2017** Rodent linifanib hypertension Hypertension in patients as rats predicted
	colorectal cancer anti-EGFR antibodies (3)	**2018** EGFR-expressing cancer cell lines and A431 xenografts (Sym004)	**2017** Resistant CRC lines Regimen reached, trial failed, program stopped	**2017** Resistant CRC lines Phase 2 OS failed in acquired resistance	**2013** Cynomolgus repeat-dose toxicology with cetuximab comparator (Sym004)
#11	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2022** Chronically HBV-infected chimpanzees	**2018** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)	**2023** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2023** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)
	chronic hepatitis B virus infection HBsAg release (NAP) inhibitors (3)	**2020** DHBV-infected Pekin ducks △ Hepatocytes miss HBsAg block	**2020** CD-1 mouse and cynomolgus PK/toxicology (uninfected)	**2016** DHBV-infected Pekin ducks	**2020** CD-1 mouse and cynomolgus PK/toxicology (uninfected)
#12	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and patient-derived xenografts in immunodeficient mice	**2018** Lenvatinib population PK and exposure-response model in HCC, choosing weight-based starting dose	**2020** Mouse HCC with diverse transposon-delivered genetic drivers, lenvatinib vs sorafenib susceptibility	**2015** GLP repeat-dose toxicology in rat plus dog or monkey
	hepatocellular carcinoma anti-PDCD1 antibodies (4)	**2025** In-vitro PD-1 receptor-occupancy titration of penpulimab on human cells	**2025** Cynomolgus monkey 6- and 13-week repeat-dose toxicology of penpulimab, no occupancy readout	**2023** Syngeneic orthotopic mouse HCC under a rat anti-mouse PD-1 surrogate, with or without apatinib	**2022** PD-1 knockout mice △ Monkeys silent; patients had irAEs
	hepatocellular carcinoma KDR (VEGFR) inhibitors (3)	**2020** Human liver-cancer cell-line xenografts in athymic rats (KRN951 panel) and nude mice (apatinib)	**2021** Human liver-cancer cell-line xenografts in athymic rats (KRN951 panel) and nude mice (apatinib)	**2021** Syngeneic murine HCC allograft in immunocompetent mice, apatinib monotherapy	**2021** GLP repeat-dose tox and cardiovascular safety pharmacology in rodent plus dog or monkey
	hepatocellular carcinoma anti-CD274 antibodies (3)	**2021** Primary human T-cell activation assay (proliferation, IFN-gamma) with durvalumab	**2022** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab	**2022** Syngeneic CT26 colorectal carcinoma, rat anti-mouse PD-L1 surrogate, alone or with oxaliplatin	**2025** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
	schizophrenia SLC6A9 inhibitors (4)	**2018** Rat CSF/striatal glycine by microdialysis △ Ketamine WM readout unmoved	**2025** Rat CSF/striatal glycine by microdialysis 10 mg hit PD target; failed	**2023** Human ketamine-challenge working-memory fMRI △ Three phase 3s flat on MCCB	**2026** Rodent haematology (haemoglobin) in pharmacology and toxicology
	schizophrenia CHRNA7 agonists (3)	**2024** Pig [11C]NS14492 PET alpha7 occupancy	**2013** Rodent screens: PPI, social behaviour, object recognition Above rat-effective dose, no effect	**2016** Rodent screens: PPI, social behaviour, object recognition ABT-126 phase 2b missed MCCB	**2011** Rodent screens: PPI, social behaviour, object recognition
#14	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors	**2026** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors △ Hypoxia-sensitisation bar missed	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)
	gastric cancer anti-ERBB2 antibodies (3)	**2017** Anti-HER2 bispecific: cell proliferation in vitro and xenografts	**2020** Cynomolgus single and repeat-dose tox (margetuximab) △ Target reached, JACOB failed	**2025** Anti-HER2 bispecific: cell proliferation in vitro and xenografts △ JACOB missed overall survival	**2020** Cynomolgus single and repeat-dose tox (margetuximab)
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2013** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology